

“We Rock the Hizzle and Stuff”

hints on how to write a nice research essay

Michele Rossi[†], Author two[‡]

Abstract—Future vehicular communication networks call for new solutions to support their capacity demands, by leveraging the potential of the millimeter-wave (mm-wave) spectrum. Mobility, in particular, poses severe challenges in their design, and as such shall be accounted for. A key question in mm-wave vehicular networks is how to optimize the trade-off between directive Data Transmission (DT) and directional Beam Training (BT), which enables it. In this paper, learning tools are investigated to optimize this trade-off. In the proposed scenario, a Base Station (BS) uses BT to establish a mm-wave directive link towards a Mobile User (MU) moving along a road. To control the BT/DT trade-off, a Partially Observable (PO) Markov Decision Process (MDP) is formulated, where the system state corresponds to the position of the MU within the road link. The goal is to maximize the number of bits delivered by the BS to the MU over the communication session, under a power constraint. The resulting optimal policies reveal that adaptive BT/DT procedures significantly outperform common-sense heuristic schemes, and that specific mobility features, such as user position estimates, can be effectively used to enhance the overall system performance and optimize the available system resources.

This is an example abstract. It is 204 words long, I would recommend that an abstract should not be longer than 250 words and some Transactions journals of the IEEE are currently putting a strict limit of 200 words. Here, you should briefly state:

- 1) the technical scenario/field of research and its timeliness/relevance in general (one sentence),
- 2) what you do in the report/paper and why it is important, how it advances the state of the art in its field (a few, e.g., max. 3, sentences),
- 3) summarize the main and best results of your study/proposal/method (one or two sentences),
- 4) (optional) how others could benefit from your results for further research, or within commercial products (one sentence).

The title and the abstract are probably the two most important parts of the paper/report. In the world we live in everyone is busy, and you typically have about one minute to catch the reader’s attention. An uninviting/ineffective title and a poor abstract may already demotivate a reviewer from reading your paper and moving your paper towards the rejection side in the decision process. A few rules of thumb follow.

In the title, you should avoid overstatements, be technical but at the same spur interest. Very bold statements like the famous “Attention is all you need”, which is the title of the paper where transformers were presented for the first time should be avoided. In general, they may irritate the reviewer. On the other hand, however, this title really captures the reader’s attention.

In that case, the authors took their risk and it went well. In general, I recommend more sober titles, grounded on technical content, statement implying that you solve “all problems” are rather bold and whp false, very rarely a single piece of work solves all problems.

In the abstract, I suggest you follow this presentation flow: 1) establish the context, 2) motivate the problem, 3) briefly describe your solution, and 4) present the main results of your work. Ideally, use one/two (short) sentence(s) for each of the previously mentioned items to keep your abstract short. Overall, this should be a short summary of the whole content of your paper, including your results.

See the abstract as a personal challenge for each of your papers. Finally, the abstract should contain the main message about your work, so that the reader will now what she/he can find even without reading it (as it is the case most of the times). The abstract is a mini-paper on its own and, as such, it is a major endeavor to write.

I suggest to write the Abstract as the very last thing. You may sketch it at the beginning, but then always finalize it at the end.

Index Terms—Self Organizing Maps, Unsupervised Learning, Optimization, Neural Networks, Recurrent Neural Networks. A list of keywords defining the tools and the scenario. I would not go beyond *six* keywords.

I. INTRODUCTION

Remark I.1. Template: You MUST use the present Latex template for the final report. This template mimics a typical IEEE template for scientific conference and journals.

Remark I.2. Paper contribution: First and foremost decide/understand on what precisely is the contribution of your paper against the state of the art. If you think you have several contributions, maybe just *focus on the most important one*. It may be that you can add one or two contributions as side topics, but in general you should focus on the most important one so as *to keep your paper focused*. As a side note: If you think you have several contributions for a single paper, you should probably invest in researching the state of the art and address the related work more thoroughly. Once you know your contribution, you are in the position of thinking of a good title.

Remark I.3. Contribution: Sometimes (often), it happens that you understand the *actual contribution and value* of your work only after having published it. As an example, one technique may find many more applications of what

[†]Department of Information Engineering, University of Padova, email: {rossi}@dei.unipd.it

[‡]Author two affiliation, email: {name.surname}@dei.unipd.it

Special thanks / acknowledgement go here.

you originally thought and your presentation in the original (published) paper is not formalized in a way to include all possible cases, not because it is difficult to do so, but just because you did not realize it in the first place. So, think a bit more about your work and do not rush too much.

Remark I.4. Title: Find a short and precise title for you paper exactly matching the content. It is worth investing time into this matter, as *the title will be that part of the paper by which it will be referenced* (in case it gets published).

Remark I.5. How to organize your writeup with Latex: I find it useful, especially for a paper with multiple authors, to split your manuscript into multiple source Latex files. This is very easily accomplished by having single *root* file, e.g., `main.tex`, where you define the article class, all the user-defined commands, the paper formatting options (e.g., one vs two columns, the page margins, the text size and the fonts, etc.). For this project, the root is called `template.tex`. From the root you then call, in sequential order, a number of other latex source files through the command `\input{filename.tex}`. It is often convenient to have one of such files for each section of your paper. This facilitates editing multiple sections in parallel and keeping your project synchronized with some versioning and revision control system such as `svn`, `github` or *overleaf* (highly recommended).

Remark I.6. Compiling trick: For each source Latex file that you call from the main, I recommend you copy the following command in the very first line

```
% !TEX root = template.tex
```

where `template.tex` is the name of the root Latex file. This command tells the Latex compiler that your main root file is `template.tex`, and it makes it possible to compile the paper from any of the sub-files. Example: imagine you are editing `introduction.tex` and have that open in a window of your preferred Latex editor. With this command, you can compile from the local `introduction.tex` window, without having to open and switch every time to the root file window and compile from it. This will save you hours.

Maximum length for the whole ML4HD report is 9 pages. Abstract, introduction and related work should take max two pages.

Recommended structure for the intro: you may use the following structure.

- **General (short) intro:** One paragraph to introduce your work, describing the scenario *at large*, its relevance, to prepare the reader to what follows and convince her/him that the paper focuses on an important setup/problem. Please keep this part short (I usually do five to eight lines), as this part is rather standard, **but** at the same time it has to be there.
- **Put the problem into perspective:** A second paragraph where you immediately delve into the specific problem

that you are tackling, starting to detail your contribution. Here, you describe the importance of such problem, providing examples (citing papers from the literature, possibly recent ones) of previous solutions attempts, and of why these failed *to provide a complete answer*. This second paragraph should not be too long, as otherwise the reader will get bored and will abandon your paper... It should be concisely written, something like 5 to 10 lines.

- **Present the paper contribution:** A third paragraph where you state what you do in the paper, this should also be concisely written. A good rule of thumb is to make it max 10/15 lines, BUT it should be there and should be CONVINCING. Here, you should state up front:

- 1) **problem:** the problem at stake,
- 2) **relevance:** the relevance and timeliness of what you propose,
- 3) **approach:** the technique/approach you use, possibly underlying its novelty, efficiency,
- 4) **value:** underline the value/novelty of your proposal referencing (recent) papers from the literature, underlining where they fail and how your proposal fills the gaps,
- 5) **applicability:** tell the reader how she/he can take advantage of your work, e.g., how your work/results can be reused/exploited to achieve further scientific, technical or practical (integrated into products?) goals.

- **Summary of contributions:** After this, you may want to provide an itemized list to summarize the paper contributions. Rule of thumb: from three to six items, from three to four lines each.

- **Closing (paper structure):** You finish up by detailing the paper structure, this should be three to four lines. It is customary to do so, although I admit it may be of little use. It usually goes like: “*This report (paper) is structured as follows. In Section II we describe the state of the art, the system and data models are respectively presented in Sections III and IV. The proposed signal processing technique is detailed in Section V and its performance evaluation is carried out in Section VI. Concluding remarks are provided in Section VII.*”

Remark I.7. Lately, I tend to write introduction plus abstract within a single page. This forces me to focus on the important messages that I want to deliver about the paper, leaving out all the “blah blah”. **Remember:** 1) *less is more*, 2) writing a compact (*snappy*) piece of technical text is much more difficult than writing lengthy stuff, with no space constraints.

II. RELATED WORK

Some hints:

- **Goal:** The goal of this section is to describe what has been done so far in *the* literature. You should focus on

and briefly describe the work done in the best papers that you have read.

- **Length:** One full column is fine but often this takes one column and a half. It is very easy to use a full page, although this may just due to your sloppiness... if you carefully go through the one page long version, you often find it possible to compact it in one column and a half. In any event, I would make this section no longer than one page, this leads to an overall *two pages* including abstract, introduction and related work. I believe this is a fair amount of space in most cases.
- **Approach:** For each paper that you review, you should comment on its contribution, on the good and important findings of such paper and also, **1)** on why these findings are not yet sufficient and **2)** how these findings are improved upon/extended by the work that you present here. At the end of the section, you may recap the main paper contributions (maybe one or two, the most important ones) and how these extend/improve upon previous work.
- **References:** please follow this *religiously*. It will help you a lot. Use the Latex `Bibtex` tool to manage the bibliography. A `Bibtex` example file, named `biblio.bib` is provided with this template.
- **Citing conference/workshop papers:** I recommend to always include the following information into the corresponding `bibitem` entry:
 - 1) author names,
 - 2) paper title,
 - 3) conference / workshop name,
 - 4) conference / workshop address,
 - 5) month,
 - 6) year.

Examples of this are: [1] [2].

- **Citing journal papers:** I recommend to always include the following information into the corresponding `bibitem` entry:
 - 1) author names,
 - 2) paper title,
 - 3) full journal name,
 - 4) volume (if available),
 - 5) number (if available),
 - 6) month,
 - 7) pages (if available),
 - 8) year.

Examples of this are: [3] [4] [5].

- **Citing books:** I recommend to always include the following information into the corresponding `bibitem` entry:
 - 1) author names,
 - 2) book title,
 - 3) editor,
 - 4) edition,

5) year.

Remark II.1. Note that some of the above fields may not be shown when you compile the Latex file, but this depends on the bibliography settings (dictated by the specific Latex style that you load at the beginning of the document). You may decide to include additional pieces of information in a given bibliographic entry, but please, **be consistent** across all the entries, i.e., always use the same fields for the same publication type. Note that some of the fields may not be available (e.g., the paper *volume*, *number* or the *pages*). In that case, you can just omit them.

III. PROCESSING PIPELINE

Remark III.1. On tailoring the paper structure to your needs: The structure recommended for the previous sections is rather standard and could work for different papers with differing technical content, the structure and the paper content from here on highly depends on the type of paper, possibilities are:

- a paper based on theoretical analysis,
- a paper based on experimental design/activity, proposing a new technique and analyzing its performance via simulation or (better) experiments,
- a paper containing mathematical analysis, practical design *and* experimental validation (which makes it a very solid contribution).

For the ML4HD course, we deal with machine learning and, in detail, with training and testing neural network architectures to perform some specific inference or classification task. Hence, the following structure and comments are specifically addressing this type of technical content.

Remark III.2. Why having this section: With this section, we start the technical description with a *high level* introduction of your work (e.g., processing pipeline). Here, you do not have to necessarily go into the technical details of every block/algorithm of your design, this will be done later as the paper develops. What I would like to see here is a high level description of the approach, i.e., which processing blocks you used, what they do (in words) and how these were combined, etc. This section should introduce the reader to your design, explain the different parts/blocks, how they interact and why. You should not delve into technical details for each block, but you should rather explain the big picture, the preprocessing pipeline (if any), the subsequent neural network architecture, and how they interact. **Besides a well written explanation, I suggest adding a nice diagram showing the full system, the constituent blocks and detail their interrelations. I consider such diagrams mandatory for your ML4HD report.**

Some writing tips: Sections, Figures and Tables are usually shortened as Sec., Fig., Tab.

- **Cross referencing:** In Latex, cross referencing is easy. You need to label an object through the

`\label{labelid}` command and referencing it where you need it through the `\ref{labelid}` command.

- **Suggestion:** I suggest to cross reference a table using `Tab.\ref{tab:tableid}`, the same holds for figures and sections, by just replacing “Tab.” with “Fig.” and “Sec.”. Of course when defining the table, you need to add a Latex command `\label{tab:tableid}` in the right place inside the table Latex environment to cross-link it, the same holds for figures, sections and equations. The tag `tab:tableid` is user defined and is the identifier that you associate with the table in question. You could call it `pippo` if you wish, but I recommend to use something like “`tab:tableid`” for tables, “`eq:eqid`” for equations, “`sec:secid`” for sections and so forth, where `tableid` has to be unique for each table in the document. This usually increases the readability of your document for other contributors. The same applies to figures and all other objects. I guess you got the idea. This will lead to a neat Latex code and will facilitate cross referencing while avoiding duplicate labels, especially in large Latex documents (think of a book for example).
- **But what about the tilde?** This is a nice trick I have learned from a friend. When you write `Tab.\ref{tab:tableid}` Latex knows that `Tab.` and the corresponding table number `\ref{tab:tableid}` must be displayed within the same line, i.e., they can never be broken across lines. This is nice and desirable I believe. I always use it for all referenced material, including citations; example: “As done in~\cite{suppa-wu-2019}.”
- **More about breaking stuff across lines** often times you have composed words, in line equations, etc. and for some reason you would like Latex to never break them across lines. Example: the Latex command `\mbox{Neural Networks (NN)}` is processed by Latex so that “Neural Networks (NN)” is never broken across lines. I use this very often, also for inline equations that I do not want Latex to split across subsequent lines.

IV. SIGNALS AND FEATURES

Being a machine learning paper, I would have here a section describing the signals you have been working on. If possible, you should describe, in order,

- 1) the measurement setup (if relevant), how input data is formatted (e.g., vectors of fixed size measured at regular sampling times),
- 2) how the signals were pre-processed, e.g., to remove noise, artifacts, fill gaps (missing points) or to re-interpolate the signals to represent them through a constant sampling rate, any normalization, filtering, etc.
- 3) after this, you should describe how *feature vectors* were obtained from the pre-processed signals. If signals are *time series* this also implies stating the segmentation / windowing strategy that you adopted (if any), to then describe how you obtained a feature vector for

each time window. Also, if you use existing feature extraction approaches, you may want to briefly describe them as well, in addition to (and before) your own (possibly new) feature extraction method(s).

So, please be accurate on how the input data is formatted and pre-processed before being fed to the subsequent neural network architecture. Last but not least, this section should also contain information on how you have split the dataset into training, validation and test sets. This will be briefly recalled in the “Results” section.

V. LEARNING FRAMEWORK

Here, you should finally describe the learning strategy / algorithm that you conceived and used to solve the problem at stake. A good diagram to exemplify how learning is carried out is often very useful. In this section, you should describe the learning model, its parameters, any optimization over a given parameter set, etc. You can organize this section into sub-sections. You are free to choose the most appropriate structure.

Remark V.1. Note that the diagram that you put here differs from that of Section III as here you show the details of how your learning framework, the core of it, is built. In Section III, you should instead provide a high-level description of the involved processing blocks, i.e., describing the *processing flow* and the rationale behind it.

On math typesetting: there are many Latex tricks that you should use to produce a high quality technical essay. A few recommendations are listed below, in random order. Some of such things (such as punctuation, use of math operators and parentheses) usually irritate me, as they denote a sloppy/careless writeup:

- **Vectors and matrices:** x is a scalar, whereas $\mathbf{x}$ (in bold) is a vector, and $\mathbf{X}$ is a matrix with elements $\mathbf{X} = [x_{ij}]$. For bold symbols you may use the `\bm` Latex command, e.g., `\bm{x}`.
- **Operators:** such as `max`, `min`, `argmax`, `argmin` and special functions such as `log(\cdot)`, `exp(\cdot)`, `sin(\cdot)`, `cos(\cdot)` are obtained through specific latex commands `\min`, `\max`, `\arg\!min`, `\arg\!max`, `\log`, `\exp`, `\sin`, `\cos`, etc. Use them! *log*, *exp*, *min*, *sin*, *cos*, etc., look ugly.
- **Sets** can be represented through calligraphic fonts, e.g., $\mathcal{S}$, $\mathcal{F}$, $\mathcal{B}$, etc., obtained using the Latex command `\mathcal{S}`, etc.
- **Equations:** for a single equation use the `equation` Latex environment. Example:

$$\sigma(x) = \frac{1}{1 + e^{-x}}. \quad (1)$$

Now, using round brackets (and) we get

$$\sigma(x) = \left(\frac{1}{1 + e^{-x}} \right), \quad (2)$$

but this looks ugly, you should use “`\left (`” for “(” and “`\right )`” for “)”, obtaining

$$\sigma(x) = \left(\frac{1}{1 + e^{-x}} \right). \quad (3)$$

- **Punctuation (important):** Displayed equations are usually considered to be part of the text and, in turn, they will get the very same punctuation as if they were inline with the text (and part of the sentence). If the sentence ends with a displayed equation, the equation gets a period “.” right after it, see Eq. (1). If the equation is instead part of a running sentence, which is continued after it, then the equation may be ended by a “,” as in Eq. (2). Use the standard grammar rules and your good sense of flow to assess how equations should be punctuated, I usually read through as if they were plain text.

VI. RESULTS

In this section, you should provide the numerical results. You are free to decide the structure of this section. As a general “rule of thumb”, use plots to describe your results, showing, e.g., precision, recall and F-measure as a function of the system (learning) parameters. You can also show the precision matrix. You can also use tables (very common in machine learning papers) to display the accuracy, precision, recall, memory size, number of parameters, etc., of competing approaches.

Remark VI.1. Present the material in a progressive and logical manner, starting with simple things and adding details and explaining more complex findings as you go. Also, do not try to explain/show multiple concepts within the same sentence. Try to **address one concept at a time**, explain it properly, and only then move on to the next one. Do not take anything for granted, the reader does not necessarily know all the concepts or tools that you use. So, it is good practice to either explain briefly (if the explanation is not too long) or provide adequate references when you introduce a concept or use a tool for the first time in the paper.

Remark VI.2. The best results are obtained by generating the graphs using a vector type file, commonly, either encapsulated postscript (eps) or pdf formats. To plot your figures, use the Latex `\includegraphics` command. Lately, I tend to use pdf more.

Remark VI.3. If your model has hyper-parameters, show selected results for several values of these. Usually, tables are a good approach to concisely visualize the performance as hyper-parameters change. It is also good to show the results for different flavors of the learning architecture, i.e., how architectural choices affect the overall performance. An example is the use of CNN only or CNN+RNN, GNN, or using inception for CNNs, dropout for better generalization or attention models. So you may obtain different models that solve the same problem, e.g., CNN, CNN+RNN, GNN, CNN+inception, CNN+attention, and so forth.

Remark VI.4. Ask yourself the reason for the results you are getting. If something weird occurs, e.g., the training accuracy is lower than the accuracy on the test set, you need to understand why. This is good practice, and through this process you may discover bugs in the code or in the way you structured your training and evaluation phases. Also, a good results section should contain useful observation, which explain the reason behind the results in the tables or in the plots that you show. Just reporting the numerical results, with no explanation of their meaning should be avoided.

VII. CONCLUDING REMARKS

This section should take max half a page. In case you have some important considerations to write, you may also extend the section to 3/4 of a page.

In many papers, here you find a summary of what done. It is basically an abstract where instead of using the present tense you use the past participle, as you refer to something that you have already developed in the previous sections. While I did it myself in the past, I now find it rather useless. I instead recommend to summarize what is done *and* also report some useful considerations that cannot be found elsewhere, these include (examples): 1) the validity/applicability and the limits of the proposed approach, 2) any possible extensions (left as a future work).

Hence, what I would like to see here is:

- 1) a short summary of what done,
- 2) some (possibly) intelligent observations on the relevance and *applicability* of your algorithms / findings,
- 3) what is still missing, and can be added in the future to extend your work.

The idea is that this section should be *useful* and not just a repetition of the abstract (just re-phrased and written using a different tense...).

Moreover: being a project report, I would also like to see a specific paragraph stating

- 4) what you have learned, and
- 5) any difficulties you may have encountered.

VIII. EXAM MODALITIES AND FINAL SCORE

What you need to do to pass the exam:

- Optional: team up with another student. Max. group size is **two students** per group;
- Identify a project to work on, devise your own neural network architecture and test it on the provided dataset;
- **Prepare a written project report** including: i) diagrams, ii) configuration pars, iii) results, iv) your discussion;
- **Prior to presenting your work:** upload i) your written report and ii) your code;

- **Present your work** using slides (max. duration is 20 minutes): take turns in presenting your work, your individual contribution to the project should clearly emerge. Optional but recommended: a final and quick demo with running code is appreciated and will be considered in the calculation of the final grade (see below).

Your final grade will be obtained taking into account the following three criteria:

- **Project (55 points):**
 - **Originality (10 pt.):** is the approach original, involving a recent / new technique or a combination of well known techniques, but carried out in a sensible manner? Also, is the project taking an interesting angle (e.g., looking at small-memory footprint schemes)?
 - **Data preprocessing techniques (10 pt.):** is the data preprocessing technique clearly explained and implemented? Did the project involve any solid data preprocessing? (e.g., MFCC feature extraction, normalization, filtering of any kind, image contrast enhancement via CLAHE, etc.) Did the project involve any data augmentation technique?
 - **Learning architectures (15 pt.):** complexity and originality of the developed learning architecture.
 - **Comparison against alternative or existing approaches (10 pt.):** are different approaches compared against one another? Example: CNN vs RNN vs GNN, etc. with and without attention, etc. Is an ablation study performed? (*An ablation study is a process of removing or disabling components of a system to understand their contribution to its overall performance. By systematically taking away parts of a model / algorithm and observing how the performance changes, researchers can assess the impact of each feature and validate design choices.*)
 - **Live demo of the code (10 pt.):** did the student(s) show/commented their code? Commenting its execution and showcasing one example case?
- **Written report (35 points):**
 - **Clarity of exposition (10 pt.):** is the project report well and clearly written? Did the student(s) duly explain the approach using diagrams and accompanying text?
 - **Completeness of results (10 pt.):** did the student(s) consider accuracy *and* also memory size and number of parameters (as proxies of the complexity of the approach)? Training and inference times can also be accounted for.
 - **Technical soundness (15 pt.):** is the approach (preprocessing, training and test phases) implemented and trained in a technically sound manner? Are there unexplained or strange results pointing at possible mistakes made in the training and/or test phases? Example: augmented data should only be used during training, and should never leak into the test phase. Also, training data should never be used for testing.

Is the code well and correctly written? Does/do the student(s) show proficiency with the developed code?

- **Oral exposition (20 points):**
 - **Duration (10 pt.):** your talk must take max. 20 minutes, using slides in a conference-style presentation.
 - **Clarity of exposition (10 pt.).**

The final grade will be computed as

$$\text{grade} = 0.427 \times \text{tot_points} - 11.69, \quad (4)$$

where tot_points corresponds to the sum of the points accrued for the criteria above.

REFERENCES

- [1] M. Zargham, A. Ribeiro, A. Ozdaglar, and A. Jadbabaie, "Accelerated dual descent for network optimization," in *American Control Conference*, (San Francisco, CA, US), June 2011.
- [2] C. M. Sadler and M. Martonosi, "Data compression algorithms for energy-constrained devices in delay tolerant networks," in *ACM SenSys*, (Boulder, CO, US), Oct. 2006.
- [3] C. E. Shannon, "A mathematical theory of communication," *The Bell System Technical Journal*, vol. 27, pp. 379–423, July 1948.
- [4] S. Boyd, N. Parikh, E. Chu, and B. P. J. Eckstein, "Distributed Optimization and Statistical Learning via the Alternating Direction Method of Multipliers," *Foundations and Trends in Machine Learning*, vol. 3, pp. 1–122, Jan. 2011.
- [5] D. Zordan, B. Martinez, I. Vilajosana, and M. Rossi, "On the Performance of Lossy Compression Schemes for Energy Constrained Sensor Networking," *ACM Transactions on Sensor Networks*, vol. 11, pp. 15:1–15:34, Aug. 2014.